

Q1: What is Object-Oriented Programming and its core principles?

Ans: Object Oriented Programming is a paradigm centered around objects rather than functions. Its four primary pillars are Encapsulation, Abstraction, Inheritance, and Polymorphism. Encapsulation bundles data and methods while restricting direct access.

Q2: Explain Polymorphism with its types.

Ans: Polymorphism allows an entity to take multiple forms. Compile-time polymorphism is achieved via method overloading. Runtime polymorphism is achieved via method overriding in subclasses.

Q3: Describe Exception Handling mechanism.

Ans: Exception handling prevents unexpected crashes. The try block encloses risky code, catch handles the error gracefully, and finally ensures cleanup operations are always completed.